

$$P_i = 1.2 \text{ N s}$$

6	C	The thrust must provide an upward force to propel the rocket upward. This force must be least at equal to its weight. Thrust = $v \left(\frac{dm}{dt} \right)$ $v \left(\frac{dm}{dt} \right) = Mg$ $\frac{dm}{dt} = \frac{Mg}{v} = \frac{(500)(9.81)}{1000} = 4.9 \text{ kg s}^{-1}$
7	C	